



An herbal formulation “Shenshuaifu Granule” alleviates cisplatin-induced nephrotoxicity by suppressing inflammation and apoptosis through inhibition of the TLR4/MyD88/NF- κ B pathway

Xiaoming Jin ^{a,1}, Riming He ^{a,1}, Jiahui Liu ^b, Yuzhi Wang ^a, Zhongtang Li ^a, Beibei Jiang ^a, Jiandong Lu ^{c,**}, Shudong Yang ^{a,*}

^a Department of Nephrology, Shenzhen Traditional Chinese Medicine Hospital, The Fourth Clinical Medical College of Guangzhou University of Chinese Medicine, Shenzhen, Guangdong, 518033, China

^b Shenzhen Traditional Chinese Medicine Hospital Affiliated to Nanjing University of Chinese Medicine, Shenzhen, Guangdong, 518033, China

^c Department of Nephrology, Shenzhen Traditional Chinese Medicine Hospital, Nanjing University of Chinese Medicine, Shenzhen, 518033, China

ARTICLE INFO

Handling Editor: Dr. Thomas Efferth

Keywords:

Cisplatin-induced nephrotoxicity
A herbal formulation
Inflammation
Apoptosis
The TLR4/MyD88/NF- κ B pathway

ABSTRACT

Ethnopharmacological relevance: Shenshuaifu Granule (SSF) is an in-hospital preparation approved by the Guangdong Food and Drug Administration of China. It has been clinically used against kidney diseases for more than 20 years with a definite curative effect.

Aim of the study: Cisplatin (CDDP) is a first-line chemotherapeutic drug in clinical practice, primarily excreted by the kidney with nephrotoxicity as a common side effect. Approximately 5–20% of cancer patients develop acute kidney injury (AKI) after chemotherapy; however, prevention and control strategies are currently unavailable. Therefore, it is important to identify safe and effective drugs that can prevent the nephrotoxicity of CDDP. SSF is an herbal formulation with 8 herbs, and has been used to protect the kidney in China. Nonetheless, its mechanism in relieving CDDP nephrotoxicity remains unclear. Therefore, this work attempt to prove that SSF can alleviate CDDP nephrotoxicity. We also explore its mechanism.

Materials and methods: First, Thin Layer Chromatography (TLC) of a few herbs in SSF were performed for quality control. Several open-access databases were used to identify the active ingredients of SSF, their corresponding targets, and CDDP-induced nephrotoxicity targets. We performed Protein-Protein Interaction (PPI), Gene Ontology (GO), and Kyoto Encyclopedia of Genes and Genomes (KEGG) enrichment analyses. Next, the results of network pharmacology were validated using CDDP-induced nephrotoxicity mouse models. Renal function in the mice was assessed by analyzing the levels of serum creatinine (Scr) and blood urea nitrogen (BUN). On the other hand, renal damage was assessed by determining the level of tubular injury and apoptotic cells using Periodic acid-Schiff (PAS) staining and Terminal DUTP Nick End-Labeling (TUNEL) staining, respectively. The expression of inflammatory and apoptotic-related targets including IL-1 β , IL-6, TNF- α , Cox-2, Bax, Bcl-2, Cleaved-caspase 3, and Cleaved-caspase 9 was determined using Western Blot (WB) and Immunohistochemistry (IHC). Furthermore, WB was used to analyze the expression of proteins associated with the TLR4/MyD88/NF- κ B pathway in the kidneys of mice with CDDP-induced nephrotoxicity. Finally, molecular docking simulations were performed to evaluate the binding abilities between major active ingredients of SSF and core targets.

Result: Through network pharmacology, we identified 127 active ingredients of SSF and their corresponding 134 targets. Additional screening identified 14 active ingredients and 17 targets for further analysis. In biological process (BP), the targets were enriched in inflammation and apoptosis, among others. In KEGG terms, they were enriched in apoptosis and NF- κ B pathways. Animal experiments revealed that SSF significantly reduced the levels of Scr and BUN and prevented renal tubular damage in mice treated with CDDP. In addition, SSF inhibited inflammation and apoptosis by targeting the TLR4/MyD88/NF- κ B pathway. Molecular docking revealed good binding capacities of active ingredients and core targets.

* Corresponding author.

** Corresponding author.

E-mail addresses: 20213184@njucm.edu.cn (J. Lu), shudong_yang@126.com, ysd0870@gzucm.edu.cn (S. Yang).

¹ These authors contributed equally to this work and should be considered co-first authors.

Conclusion: In summary, the experimental findings were consistent with the network pharmacological predictions. SSF can inhibit inflammation and apoptosis by targeting the TLR4/MyD88/NF-κB pathway. Taken together, our data suggest that SSF is an alternative agent for the treatment of CDDP-induced nephrotoxicity.

Abbreviations:

SSF	Shenshuaifu Granule
CDDP	Cis-diamine dichloroplatinum, Cisplatin
AKI	Acute kidney injury
TLC	Thin Layer Chromatography
PPI	Protein-Protein Interaction
BC	Betweenness Centrality
CC	Closeness Centrality
DC	Degree Centrality
GO	Gene Ontology
KEGG	Kyoto Encyclopedia of Genes and Genomes
BP	Biological Process
CC	Cellular Component
MF	Molecular Function
Scr	Serum creatinine
BUN	Blood urea nitrogen
PAS	Periodic acid-Schiff

TUNEL	Terminal Dntp Nick End-Labeling
WB	Western Blot
IHC	Immunohistochemistry
TNF-α	Tumor necrosis factor-α
IL-6	Interleukin-6
IL-1β	Interleukin-1β
Cox-2	Cyclooxygenase-2
Bcl-2	B-cell lymphoma-2
Bax	Bcl-2-associated X
CD 14	Cluster of differentiation 14
TLR4	Toll-like receptor 4
MyD88	Myeloid differentiation factor 88
IKK	Inhibitor of kappa B kinase
P-IKK	Phospho-inhibitor of kappa B kinase
IκB	Inhibitor of nuclear factor kappa-B
P-IκB	Phospho-Inhibitor of nuclear factor kappa-B
NF-κB p65	Nuclear factor kappa-B p65
P-NF-κB p65	Phospho-Nuclear factor kappa-B p65

1. Introduction

Cisplatin, also called cis-diamine dichloroplatinum (CDDP), was discovered in 1965 by Rosenberg, and in 1978, it became the first FDA-approved chemotherapeutic drug. Due to its excellent effect against several solid cancers, it is still a first-line drug for the treatment of cancers (Dasari and Bernard Tchounwou, 2014). The efficacy of CDDP is proportional to its dose (Santos et al., 2020); however, high doses are associated with severe toxic side effects, which significantly hinders its clinical application (Qi et al., 2019). In clinical practice, nephrotoxicity of CDDP is a common cause of acute kidney injury (AKI). AKI is one of the most common complications associated with cancer patients receiving CDDP treatment. Since the kidney is a vital organ for the excretion of CDDP (Ozkok and Edelstein, 2014), a single dose of CDDP (50–100 mg/m²) may cause AKI in about 1/3 patients, potentially increasing the risk of death in ICU patients by 50% (Ramesh and Reeves, 2004). Hydration, supplementation of magnesium, and forced diuresis are the major therapeutic interventions for AKI; nevertheless, these interventions may increase the risk of dehydration and electrolyte abnormalities (Crona et al., 2017). Therefore, there is an urgent need for safe and effective drugs to alleviate CDDP-induced nephrotoxicity.

The mechanism of CDDP-induced nephrotoxicity includes oxidative stress, inflammatory response, apoptosis, and mitochondrial dysfunction (Galgamuwa et al., 2016). Traditional Chinese medicine (TCM) believes that the pathogenesis of CDDP-induced nephrotoxicity is qi deficiency of spleen and kidney, combined with drug toxicity, and the accumulation of blood stasis and turbidity toxin. In recent years, Chinese herbal formulations have garnered increased attention and have been extensively studied in the treatment of various diseases. Research confirms a traditional Chinese herbal formulation called Jianpi Yishen Tang (JPYST) attenuates CDDP-induced AKI in mice through its anti-apoptotic and anti-inflammatory effects (He et al., 2022).

Shenshuaifu Granule (SSF) is an in-hospital preparation of Shenzhen Traditional Chinese Medicine Hospital approved by the Guangdong Food and Drug Administration of China. It comprises *Astragalus membranaceus* (Fisch.) Bge., *Alisma orientale* (Sam.) Juzep., *Ilex asprella*

(Hook. et Am.) Champ. ex Benth, *Salvia miltiorrhiza* Bge., *Poria cocos* (Schw.) Wolf, *Lycium barbarum* L., *Leonurus japonicus* Houtt., *Rheum officinale* Baill. (Table 1). SSF can nourish qi and yin, activate blood circulation to remove blood stasis, clear heat and detoxification. Clinical observations confirmed its efficacy in the treatment of chronic kidney injury (CKD). Moreover, no abnormal changes in liver function have been noted during medication (Yang et al., 2012; Zhang et al., 2020). However, the therapeutic effect and mechanism of SSF on CDDP-induced nephrotoxicity remain unclear. Network pharmacology is a new discipline of multi-target drug research based on systems biology (Gu et al., 2022). Molecular docking technology can predict the relationship between active ingredients and related targets in herbal medicines by simulating and calculating the interaction force between molecules of different structures (Singh et al., 2022). The

Table 1

The Latin name, English name, Chinese name, and grams of herbs in SSF.

Latin Name ^a	English Name ^a	Chinese Name	Medicinal Part	Grams (g) ^b
<i>Astragalus membranaceus</i> (Fisch.) Bge.	Radix Astragali	Huang Qi	Root	25
<i>Alisma orientale</i> (Sam.) Juzep.	Alismatis Rhizoma	Ze Xie	Tuber	15
<i>Ilex asprella</i> (Hook. et Am.) Champ. ex Benth	Radix Ilicis Asprellae	Gang Mei Gen	Root	15
<i>Salvia miltiorrhiza</i> Bge.	Salvia Miltiorrhiza	Dan Shen	Root; Rhizome	7.5
<i>Poria cocos</i> (Schw.) Wolf	Poria	Fu Ling	Sclerotium	7.5
<i>Lycium barbarum</i> L.	Lycii Fructus	Gou Qi zi	Fruit	7.5
<i>Leonurus japonicus</i> Houtt.	Leonuri Herba	Yi Mu Cao	The aboveground section	7.5
<i>Rheum officinale</i> Baill.	Rhubarb	Da Huang	Root; Rhizome	5

^a The Latin name and English name of herbs refer to the World Checklist of Selected Plant Families (<https://wcp.science.kew.org>).

^b The Grams of various herbs contained in each dose of SSF.

comprehensiveness, integrity, and systematisms of network pharmacology are consistent with the characteristic of “Jun (Monarch)-Chen (Minister)-Zuo (Assistant)-Shi (Courier)” in Chinese herbal formulations (Zhang et al., 2019a), which provides a new perspective for the mechanism study of SSF. In this study, we used network pharmacology to predict the mechanism of SSF in relieving CDDP-induced nephrotoxicity. Our findings were verified using animal experiments.

2. Materials and methods

2.1. Thin Layer Chromatogram (TLC) of SSF

2.1.1. The testing solutions and reference standards

Exactly 1 g of SSF was added to 5 mL of methanol, and after shaking via sonication (120 W, 45 kHz) for 30 min, activated carbon was added to decolorize and filtered, which was used as the test solutions. A similar method was used to prepare the Rhubarb reference drug (0.2 mg/mL). We accurately weighed appropriate amount of Astragaloside IV (Batch number:110781-201717, the Institute for Food and Drug Control of China, China), Salvianic acid A (Batch number:110855-201915, the Institute for Food and Drug Control of China, China), and Rhein (Batch number: 110757-201607, the Institute for Food and Drug Control of China, China), respectively, and obtained the reference standards 0.7 mg/mL of Salvianic acid A sodium, 1 mg/mL of Astragaloside IV and 1.5 mg/mL of Rhein after adding ethanol to dissolve.

2.1.2. Determination of Radix Astragali, Salvia Miltiorrhiza, and Rhubarb in SSF

Under the condition of the temperature of 20°C and humidity of 50%, we performed the experiments on Radix Astragali, Salvia Miltiorrhiza, and Rhubarb in SSF, respectively. A certain amount of testing solutions and reference standards were dabbed on the same silica gel GF254 thin layer plate, respectively, and developed them for 15 cm with the developing agent (ethyl acetate: glacial acetic acid: methanol: water=5.5:1:0.2:1). After drying, they were observed under ultraviolet light.

2.2. Prediction of network pharmacology

2.2.1. Acquisition of active ingredients and targets related to SSF

The TCMSP database (<https://old.tcmsp-e.com/tcmsp.php>) was used to identify the active ingredients of Radix Astragali, Salvia Miltiorrhiza, Rhubarb, Lycii Fructus, Leonuri Herba, Alismatis Rhizoma, and Poria and their corresponding targets (Ru et al., 2014). We set the Oral Bioavailability (OB) as $\geq 30\%$ and Drug Likeness (DL) as ≥ 0.18 . Given that data on Radix Illicis Asprellae in the TCMSP database were unavailable, its active ingredients were identified by searching the literature (Chen et al., 2019). Potential active ingredients of Radix Illicis Asprellae were imported into the Swiss Target Prediction database (<http://swisstargetprediction.ch/>) with “Homo sapiens” as the research species, the predicted targets were obtained by screening with Probability ≥ 0.5 (Daina et al., 2019).

2.2.2. Acquisition of targets related to CDDP-induced nephrotoxicity

The keywords “CDDP-induced nephrotoxicity” or “CDDP-induced acute kidney injury” were used to search the GeneCards database (<https://www.genecards.org>) and OMIM database (<https://www.omim.org/>) for CDDP-induced nephrotoxicity-related targets (Safran et al., 2021).

2.2.3. Herbs–active ingredients–targets interaction and acquisition of main active ingredients

The Venny 2.1 (Venny 2.1.0 (csic.es)) was used to identify common targets of SSF and CDDP-induced nephrotoxicity. These common targets were considered the targets of SSF in alleviation of CDDP-induced nephrotoxicity. Cytoscape 3.9.0 was used to draw the network diagram of “Herbs-active ingredients-targets interaction”. “nodes” in the

network diagram represented the herbs, active ingredients, and targets, whereas the “edges” represented the relationship among the nodes.

2.2.4. PPI and acquisition of core targets

Protein-Protein Interaction (PPI) of the common targets of SSF and CDDP-induced nephrotoxicity was constructed using the STRING platform (<https://cn.string-db.org/>) with the species set to “Homo sapiens” (Szkarczyk et al., 2021). Then the median of Betweenness Centrality (BC), Closeness Centrality (CC), and Degree Centrality (DC) of PPI were analyzed through network topology analysis function of Cytoscape 3.9.0 to obtain the core targets.

2.2.5. GO and KEGG enrichment analysis

GO and KEGG enrichment analysis were performed using the David database (<https://david.ncicrf.gov/>) with the species set to “Homo sapiens” (Huang et al., 2009). The results were observed using the Weishengxin software (<http://www.bioinformatics.com.cn/>).

2.2.6. Molecular docking of main active ingredients and core Targets

The PubChem IDs of the 14 main active ingredients were obtained from the PubChem database (<https://pubchem.ncbi.nlm.nih.gov/>) (Kim et al., 2020). The Chem 3D software was used to optimize energy minimization, whereas Auto Dock (<https://autodock.scripps.edu/>) was used to handle molecular cyclic peptides. The 3D protein structures of the core targets were downloaded from the RCSB PDB (<https://www.rcsb.org/>) followed by the removal of the water molecule and original small molecule ligand of the receptor protein, and the addition of non-polar hydrogen and electric charge. Auto Dock Vina (<https://vina.scripps.edu/>) was used to perform molecular docking simulations and the affinity between the main active ingredients and their targets was evaluated using the binding energy index.

2.3. Validation of animal experiments

2.3.1. Drug

Shenshuaiyu granule was manufactured by Shenzhen Traditional Chinese Medicine Hospital (Shenzhen, China) and approved by the Guangdong Food and Drug Administration of China. (Executive standard: Guangdong Province ZB20111557. Approval number: Yueyaozhizhi Z20070565. Production batch:20210901). Specification: Each gram of SSF is equivalent to 4.5 g of herbs. Each dose of SSF weighs 20 g. Preparation method: The herbs (Table 1) in SSF were identified by Dr. Jianping Chen from Shenzhen Traditional Chinese Medicine Hospital. The voucher specimens were kept at Shenzhen Traditional Chinese Medicine Hospital. A total of 8 types of herbs (4500 g) were decocted twice at a fixed proportion for 2 h each time, filtered, mixed, and filtered. The filtrate was concentrated into a thick extract with a relative density of 1.25–1.30 (80°C). After the filtrate cooled down, 1000 ml ethanol was added, stirred, kept for 24 h, filtered, recovered ethanol, and concentrated to form a thick extract with a relative density of 1.25–1.30 (80°C). We added a quantity of sucrose and dextrin (3:1) to the thick extract. Finally, it was well mixed, made granules (1000 g), and dried at 65–75°C. Quality Control of SSF: The description, particle size, moisture, dispersion, weight variation, and microbial limit of SSF were inspected based on the general requirements for granules 0104, Volume IV, Chinese Pharmacopoeia (2020 edition). At the same time, the TLC of SSF detected the spots with the same color and position corresponding to the Astragaloside IV reference solution, the Salvianic acid A sodium, Rhein reference solution and Rhubarb reference drug solution.

2.3.2. Animal model

Healthy male C57/BL/6J mice, weighing 20–22 g and aged 7–8 weeks, were purchased from the Guangdong Provincial Medical Laboratory Animal Center. The mice were acclimatized to the laboratory environment under standard laboratory conditions with light/dark cycles, and had free access to water and common chow. After 1 week, 18

mice were randomly divided into the normal group, the CDDP group, and the CDDP+SSF group, with 6 mice in each group. The mice in the CDDP group and the CDDP+SSF group were intraperitoneally injected with 20 mg/kg CDDP (Sigma Aldrich, USA) (Hamano et al., 2021), whereas those in the control group received an equal volume of distilled water. The mice in the CDDP+SSF group were pretreated with 5.2 g/kg SSF via intragastric gavage for 5 consecutive days before CDDP injection. The dosage and method of administration of mice were based on the following: The oral dose of SSF was 40 g every day for adults that the standard body weight of 70 kg. According to the methodology of the pharmacological test (Nair et al., 2018), the adult dose was converted to the mouse dose based on body surface area, if the human dose is X mg/kg, the dose for mouse = $X \text{ mg/kg} \times 70 \text{ kg} \times 0.0026 / 20 \text{ g}$. Therefore, the gavage dose of SSF in mice was 5.2 g/kg/d. After dissolving 52 g of SSF granule in 1000 mL of physiological saline, and configuring it into the solution of SSF with a concentration of 52 mg/mL, the gavage intervention was performed based on the actual body weight of the mice. The mice were anesthetized with isoflurane gas 72 h after CDDP injection, before being euthanized to obtain venous blood and kidney specimens, and the specimens were immediately frozen at -80°C for use (Fig. 5A). All animal experimental protocols were reviewed by the animal ethics committee of Shenzhen TopBiotech Company (Approval number: TOP-IACUC-2021-0118).

2.3.3. Analysis of renal function

The levels of Serum creatinine (Scr) and blood urea nitrogen (BUN) were measured using enzymatic methods with the automatic biochemical analyzer (Hitachi Cobas 8000, Roche Diagnostics, Germany).

2.3.4. Periodic acid-schiff (PAS) assay

The kidney tissue was fixed in 10% formalin, embedded in paraffin, and then sectioned into sections of about 3–4 μm thickness. PAS staining was then performed according to the instruction of the PAS kit (Sol-aibao, China). Briefly, the paraffin sections were first placed in 60°C ovens for 1 h, followed by xylene I, II, III for 10 min each; absolute ethanol I, II, 95%, 85%, and 75% ethanol for 5 min each; and finally dewaxed in water. Subsequently, the sections were treated with PAS staining, rinsed with running water for 5 min, and sequentially placed in 95% ethanol, anhydrous ethanol II, I, xylene III, II, I, dehydrated, and sealed with neutral gum.

2.3.5. Terminal Dntp Nick End-Labeling (TUNEL) assay

TUNEL staining was performed according to the instructions of the TUNEL kit (Beyotime Biotechnology, China). Briefly, paraffin sections were placed in a 60°C oven for 1 h and dewaxed in water. The sections were then incubated with Proteinase K at room temperature for 1 h, rinsed 3 times with PBS, treated with hydrogen peroxide for 20 min, and reacted with TUNEL reaction mixture at room temperature for 1 h. The BAD chromogenic reagent was used to develop the color and the nuclei were stained using hematoxylin (Solarbio, China). The steps of dewaxing, dehydration, and mounting were similar to that of PAS staining.

2.3.6. Western Blot (WB) assay

Kidney tissues were homogenized in RIPA buffer supplemented with protease inhibitors and phosphatase inhibitors. Total protein was obtained from the homogenate after centrifugation for 10 min at 4°C , 12000 rpm. The protein concentration was measured using BCA analysis then the concentration of all the samples was adjusted to the same concentration. The protein samples were mixed with loading a buffer and boiled for 5 min. Based on the molecular weight of the target protein, 8%, 10%, or 15% gels were used to separate the proteins after loading similar amount of protein. Electrophoresis was performed for 1h, and then the proteins were transferred to the PVDF membrane using semi-dry transfer. The PVDF membranes were blocked using 5% skim milk for 1 h, and then incubated with primary antibody overnight at 4°C . The following primary antibodies were used: Anti-TLR4 Rabbit mAb,

Anti-MyD88 Rabbit mAb, Anti-Phospho-NF- κB p65 Rabbit mAb, Anti-NF- κB p65 Rabbit mAb, Anti-Phospho-IKK α/β Rabbit mAb, Anti-IKK α/β Rabbit mAb, Anti- Phospho-I $\kappa\text{B}\alpha$ Rabbit mAb, Anti-I $\kappa\text{B}\alpha$ Mouse mAb, Anti-Bax Rabbit mAb, Anti-Cleaved-Caspase 3 Rabbit mAb, Anti-Caspase 3 Rabbit mAb, and Anti-Caspase 9 Rabbit mAb (1:1000, CST, USA), Anti-Bcl-2 Rabbit mAb (1:1000, Proteintech, USA), Anti-GAPDH Rabbit mAb and Anti-GAPDH Mouse mAb (1:2000, CST, USA), Anti-TNF- α Mouse mAb, Anti-IL-6 Mouse mAb, Anti-Cox-2 Mouse mAb and Anti-IL-1 β Mouse mAb (1:400, Santa Cruz, USA). Anti-CD 14 Mouse mAb (1:1000 Abcam, USA). The PVDF membranes were then incubated with horseradish peroxidase (HRP)-conjugated secondary antibody (1:2000, Abcam, USA) in 5% skim milk for 1h. The protein bands were detected using HRP chemiluminescent substrate (Millipore, China), and the intensity of the bands was determined using Image J software.

2.3.7. Immunohistochemistry (IHC) assay

The paraffin sections were placed in a 60°C oven for 1 h, dewaxed in water, boiled with sodium citrate solution pH=6.0 for 25 min for antigen retrieval, and then cooled naturally. Sodium hydroxide solution (3%) was then added to the sections dropwise and incubated for 20 min, followed by the addition of 10% goat serum dropwise and incubation for 30 min to block the sections. Primary antibodies were then added and incubated at 4°C for 12 h. The following primary antibodies were used: Anti-Cleaved-caspase 3 Rabbit mAb (1:2000, CST, USA), and Anti-TNF- α Mouse mAb, Anti-IL-1 β Mouse mAb (1:500, Santa Cruz, USA). The tissue sections were then gently rinsed 3 times with PBS for 3 min each. SignalStain Boos reagent (CST, USA) was added dropwise and incubated for 15 min. The BAD chromogenic reagent was used to develop the color and the nuclei were stained with hematoxylin (Solarbio, China), and finally dehydrated and mounted. The steps of dewaxing, dehydration, and mounting were similar to those of PAS staining.

2.3.8. Statistical analysis

Each experiment was conducted at least 3 times, and the results were expressed as mean $\pm$ standard error (SEM). Differences among groups were evaluated using a one-way analysis of variance (Anova). Significance was set at $P < 0.05$. GraphPad Prism 9.0 was used for statistical analysis and presentation of results.

3. Results

3.1. Thin Layer Chromatogram of SSF

Fig. 1 shows the test article chromatogram shows spots of the same color at the corresponding position as the reference standards chromatogram. This is consistent with our previous quality standard study of SSF (Li and Liu, 2011), illustrating that the quality of SSF is stable.

3.2. Network pharmacology

3.2.1. Herbs- active ingredients-targets interaction

SSF contained 127 active ingredients, including 58 in *Salvia Mil-tiorrhiza*, 32 in *Lycii Fructus*, 16 in *Radix Astragali*, 8 in *Leonuri Herba*, 7 in *Rhubarb*, 6 in *Alismatis Rhizoma*, 3 in *Poria* and 3 in *Radix Ilicis Asprellae*. In total, 5 active ingredients were present in 2 or more herbs. The 127 active ingredients in SSF contained 278 targets whereas CDDP-induced nephrotoxicity had 1434 related targets. A total of 134 targets were common targets of the active ingredients and CDDP-induced nephrotoxicity and were considered SSF targets in alleviation of CDDP-induced nephrotoxicity (Fig. 2A).

3.2.2. Acquisition of main active ingredients

The average degree of connection, betweenness, and tightness between the active ingredients and the targets were 6.87, 572.46, and 0.32, respectively, based on the Cytoscape 3.9.0. The connection degree, intermediate degree, and tightness value of 14 active ingredients such as

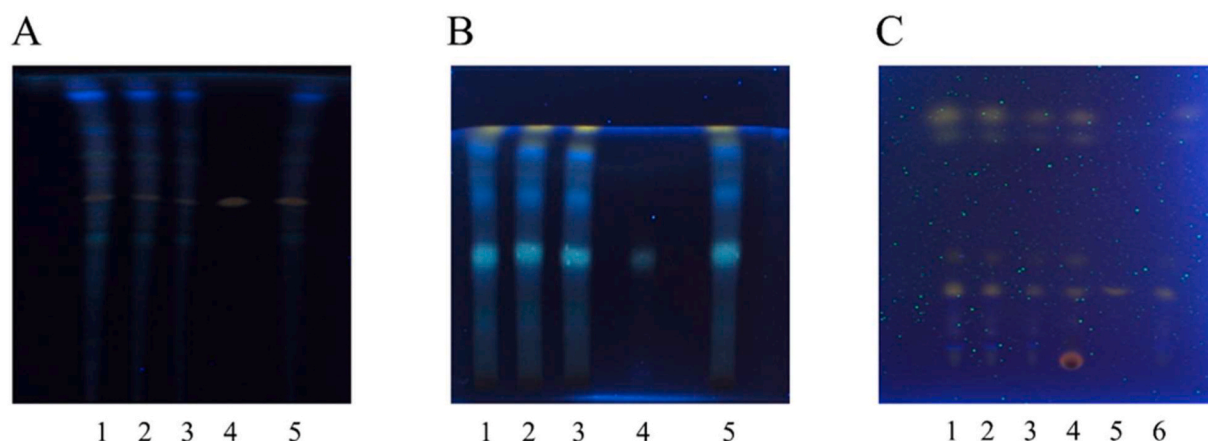


Fig. 1. The TLC of SSF (A) Chromatogram of Radix Astragali (1~2: SSF 10 μ l; 3: SSF 5 μ l; 4: Astragaloside IV 10 μ l; 5: SSF 5 μ l+Astragaloside IV 5 μ l). (B) Chromatogram of Salvia Miltiorrhiza (1~3: SSF 5 μ l; 4: Salvianic acid A sodium 5 μ l; 5: SSF 5 μ l+Salvianic acid A sodium 5 μ l). (C) Chromatogram of Rhubarb (1~3: SSF 10 μ l; 4: Rhubarb reference drug 10 μ l; 5: Rhein 10 μ l; 6: SSF 5 μ l+Rhein 5 μ l).

quercetin and luteolin were greater than or equal to their average values, suggesting that these may be the main active ingredients of SSF alleviating CDDP-induced nephrotoxicity. Table 2 shows the chemical formulas, classifications, and sources of the 14 main active ingredients. The chemical structures are shown in Fig. 2B.

3.2.3. PPI and core targets

The 134 common targets of SSF and CDDP-induced nephrotoxicity were imported into STRING software, the free nodes removed, and the PPI network constructed (Fig. 3A). The median DC, BC, and CC in the PPI were 38.5, 38.64, 0.57 as calculated using Cytoscape 3.9.0. According to the network pharmacology screening method, the targets which DC>2 times the median, BC, and CC were greater than their respective medians are core targets, which are AKT1, TNF, CASP3, TP53, VEGFA, STAT3, IL6, MYC, JUN, EGF, PTEN, PTGS2, CCND1, IL1 β , EGFR, ESR and HIF1A (Fig. 3B). However, other targets other than these 17 core targets may still play important roles. Therefore, additional targets were validated other than the 17 core targets, including Bcl-2, Caspase 9, and NF- κ B.

3.2.4. GO and KEGG enrichment analysis

The 134 common targets of SSF and CDDP-induced nephrotoxicity were inputted into the David database for GO and KEGG enrichment analysis. After setting $P < 0.01$, 86 GO and 166 KEGG were obtained. The top 10 items of BP, CC, and MF were subjected to GO enrichment analysis. In BP terms, the targets were significantly enriched in response to drugs, apoptotic process, inflammatory response, and aging among others (Fig. 4A). Among them, IL-6, IL-1 β , and other targets were enriched in the inflammatory response. CASP 3, Bcl-2, TP53, and other targets were enriched in the apoptotic process (Fig. 4C). In CC terms, the targets were significantly enriched in the nucleus, cytosol, and cytoplasm among others. In the MF term, the targets were significantly enriched in protein binding and enzyme binding among others (Fig. 4A). Theoretically, the 166 KEGG pathways obtained may all be involved in the process of SSF alleviating CDDP-induced nephrotoxicity. However, after reviewing the relevant literature, some pathways may not very meaningful for CDDP-induced nephrotoxicity, including pathways in cancer, and Epstein-Barr virus infection. Therefore, we only list 30 pathways strongly associated with CDDP-induced nephrotoxicity including TNF, Apoptosis, IL-17, and NF- κ B pathway (Fig. 4B). Among them, NF- κ B, I κ B, TNF, and other targets were enriched in the NF- κ B pathway (Fig. 4D).

3.3. Experimental verification

3.3.1. SSF protected renal tubular and renal function in mice with CDDP-induced nephrotoxicity

CDDP is mainly metabolized by renal tubules (Xu et al., 2015). Fig. 5B shows that the renal tubules of mice in the normal group were intact, while renal tubular damage characterized by tubular dilation, loss of brush borders, and vacuolization was observed in mice exposed to 20 mg/kg CDDP. SSF pretreatment significantly alleviated the manifestations of tubular damage, as indicated by the tubular injury score (Fig. 5D). Levels of Scr and BUN are indicators for assessing renal function. The Scr levels of mice exposed to 20 mg/kg CDDP ($234 \mu\text{mol/L} \pm 85.73$) were 16.71 times ($P < 0.0001$) higher than the Scr levels of normal mice ($14 \mu\text{mol/L} \pm 1.83$), whereas the Scr levels of mice pretreated with SSF ($90.33 \mu\text{mol/L} \pm 66.60$) were 3.51 times lower than the Scr levels of mice exposed to 20 mg/kg CDDP ($P < 0.01$). Similarly, the BUN levels of mice exposed to 20 mg/kg CDDP ($75.87 \text{ mmol/L} \pm 15.14$) were 6.78 times ($P < 0.0001$) higher than the BUN levels of the normal group ($11.19 \text{ mmol/L} \pm 1.32$), whereas the BUN levels of mice pretreated with SSF ($41.04 \text{ mmol/L} \pm 15.30$) were 1.85 times lower than the BUN levels in mice exposed to 20 mg/kg CDDP ($P < 0.01$) (Fig. 5C). These findings indicate that SSF significantly inhibits CDDP-induced elevation of Scr and BUN levels in mice, and effectively reduces pathological damage to the kidneys.

3.3.2. SSF reduced inflammation in mice with CDDP-induced nephrotoxicity

Inflammation is an important factor in the pathogenesis of CDDP-induced nephrotoxicity (Yang et al., 2015). Using the PPI network, TNF- α , IL-6, and IL-1 β were identified as important inflammatory factors involved in the pathogenesis of CDDP-induced nephrotoxicity. WB results revealed that the expression levels of TNF- α , IL-6, IL-1 β , and Cox-2 in mice exposed to 20 mg/kg were significantly higher than in the normal group, whereas the expression levels of these targets were significantly lower in mice pretreated with SSF than mice exposed to CDDP (Fig. 6D). These results were consistent with the results of quantification of IL-1 β and TNF- α using IHC (Fig. 6C).

3.3.3. SSF reduced apoptosis in mice with CDDP-induced nephrotoxicity

Apoptosis is an important cause of CDDP-induced nephrotoxicity (Vijayan, 2021). The TUNEL staining was used to compare the number of apoptotic cells among the three groups of mice. The number of apoptotic cells observed in the kidneys of mice in the normal group was negligible, whereas the number was significantly higher in the renal tubular cells of mice exposed to 20 mg/kg CDDP. However, SSF

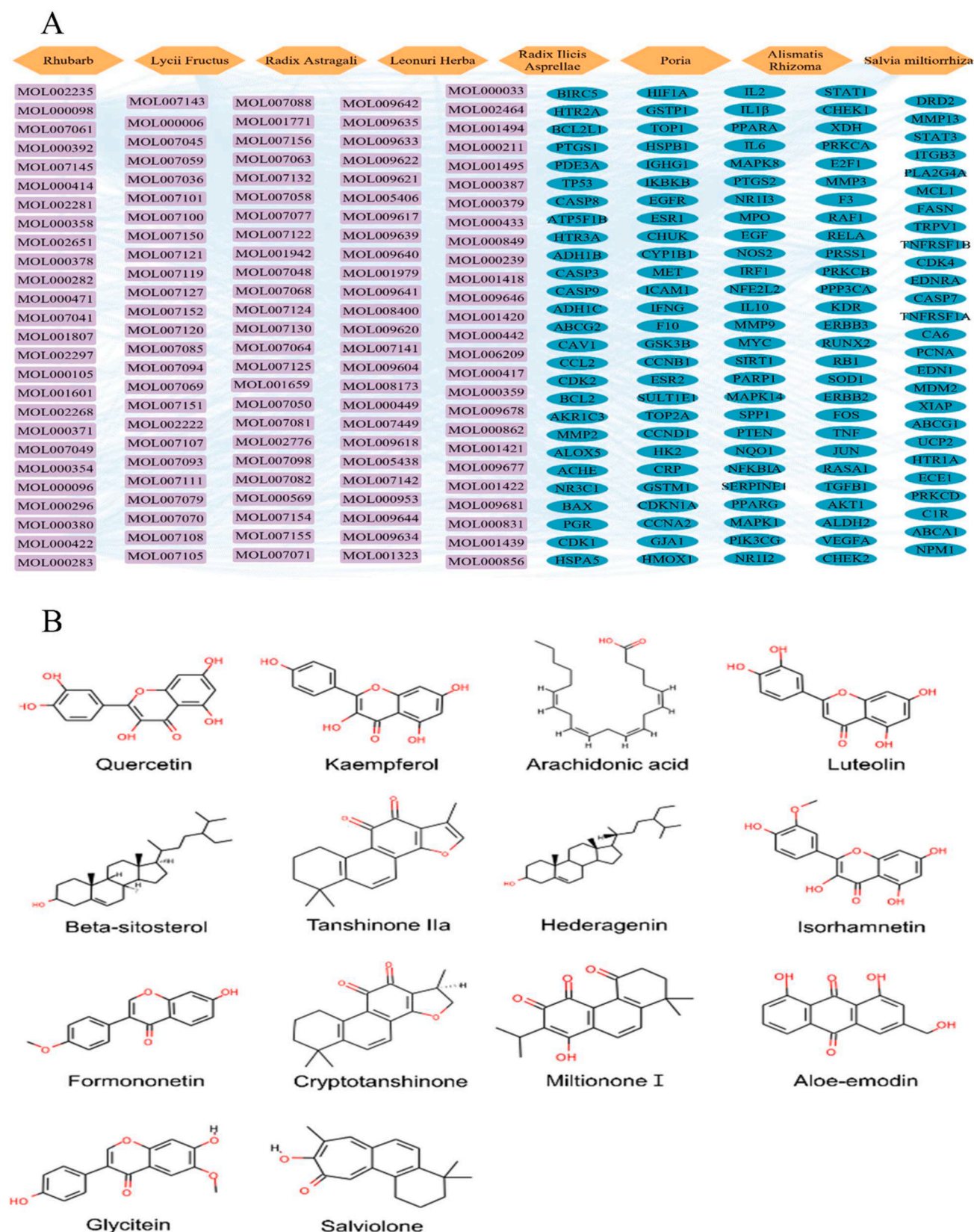


Fig. 2. The active ingredients and targets of SSF in the alleviation of CDDP-induced nephrotoxicity. (A) Herbs-active ingredients-targets interaction diagram. The orange nodes represent the herbs of SSF; The pink nodes represent the active ingredients of SSF; The blue nodes represent the common targets. (B) The chemical structures of 14 main active ingredients of SSF. (For interpretation of the references to color in this figure legend, the reader is referred to the Web version of this article.)

Table 2
The chemical formulas, classifications, and sources of 14 main active ingredients of SSF in alleviating cisplatin-induced nephrotoxicity.

Molecular ID	Chemical Name	Chemical Formula	Classification	Sources
MOL000098	Quercetin	C ₁₅ H ₁₀ O ₇	Flavones	Radix Astragali; Lycii Fructus; Leonuri Herba
MOL000422	Kaempferol	C ₁₅ H ₁₀ O ₆	Flavones	Radix Astragali; Leonuri Herba
MOL001439	Arachidonic acid	C ₂₀ H ₃₂ O ₂	Organic acid	Leonuri Herba
MOL000006	Luteolin	C ₁₅ H ₁₀ O ₆	Flavones	Salvia Miltiorrhiza
MOL000358	Beta-sitosterol	C ₂₉ H ₅₀ O	Tetracyclic triterpenoids	Lycii Fructus; Rhubarb
MOL007154	Tanshinone Ila	C ₁₉ H ₁₈ O ₃	Phenanthraquin compounds	Salvia Miltiorrhiza
MOL000296	Hederagenin	C ₃₀ H ₄₈ O ₄	Pentacyclic triterpenoids	Radix Astragali; Poria
MOL000354	Isorhamnetin	C ₁₆ H ₁₂ O ₇	Flavones	Radix Astragali; Leonuri Herba
MOL000392	Formononetin	C ₁₆ H ₁₂ O ₄	Isoflavones	Radix Astragali
MOL007088	Cryptotanshinone	C ₁₉ H ₂₀ O ₃	Diterpenoid quinones	Salvia Miltiorrhiza
MOL007119	Miltionone I	C ₁₉ H ₂₀ O ₄	Diterpenoid quinones	Salvia Miltiorrhiza
MOL000471	Aloe-emodin	C ₁₅ H ₁₀ O ₅	Anthraquinones	Radix Astragali
MOL008400	Glycitein	C ₁₆ H ₁₂ O ₅	Isoflavones	Lycii Fructus
MOL007145	Salviolone	C ₁₈ H ₂₀ O ₂	Diterpenoids	Salvia Miltiorrhiza

pretreatment significantly reduced the number of apoptotic cells in the renal tubular cells (Fig. 7A). The percentage of renal tubular apoptosis in renal tubules validated it (Fig. 7C).

Further, apoptosis was determined by analyzing the expression of Bax, Bcl-2, Cleaved caspase 3, 9, and Caspase 3, 9 using WB. As shown in Fig. 7B, the levels of Bax, Cleaved-caspase 3/Caspase 3 and Cleaved-

caspase 9/Caspase 9 in mice exposed to 20 mg/kg CDDP were significantly higher than that in normal mice, whereas pretreatment with SSF significantly reversed these expressions. Unlike the normal mice, Bcl-2 expression was downregulated in mice injected with CDDP, and SSF effectively reversed this trend. At the same time, the differences in the expression of Cleaved-caspase 3 in the kidneys of the three groups of mice were also demonstrated by IHC (Fig. 7A).

3.3.4. SSF inhibited the TLR4/MyD88/NF-κB pathway in mice with CDDP-induced nephrotoxicity

The NF-κB pathway is a classical pathway related to inflammation and apoptosis. Among the 17 main active ingredients of SSF, Quercetin, Tanshinone Ila, and Formononetin can reduce the inflammatory response in mice with CDDP-induced nephrotoxicity by inhibiting NF-κB pathway (Dou et al., 2022; Hao et al., 2021; Tan et al., 2020). Therefore, The NF-κB pathway and its upstream were validated by WB. CD 14, TLR4, and MyD88 are the upstream signals of the NF-κB pathway. In contrast with the normal group, CD 14, TLR4, and MyD88 were activated in mice exposed to 20 mg/kg CDDP. In addition, the phosphorylation levels of IKKα/β, IκBα, and NF-κB p65 were higher in mice exposed to CDDP compared to normal mice. SSF significantly suppressed the above expression compared with mice in the CDDP group (Fig. 8).

3.4. Molecular docking of main active ingredients and core targets

To identify the role of the 14 main active ingredients of SSF in inhibiting apoptosis and inflammation, molecular docking simulation experiments was conducted using the 14 main active ingredients as ligands, and the experimentally verified TNF-α, IL-6, IL-1β, Caspase 3, Caspase 9, and NF-κB p65 as receptors. From the simulations, 94 groups of binding energies were obtained and used to evaluate the binding capacity between receptors and ligands (Fig. 9A). The lower the binding energy, the more stable the conformation of binding between ligands and receptors. If the binding energy is ≤-5.0kcal/mol, it indicates that the ligand and the receptor can bind; if the binding energy is ≤-7.0kcal/mol, it indicates that there is a good affinity between the ligand and the receptor (Bailly and Vergoten, 2020). As shown in Fig.9A, the binding energies of the 14 main active compounds and 17 core targets were all ≤-5.0kcal/mol, and most of the binding energies were ≤-7.0kcal/mol, which further indicated that these 14 main active ingredients may exert anti-inflammatory and anti-apoptotic effects. Additionally, we showed the simulated docking effects of quercetin and TNF-α, Caspase 3, NF-κB

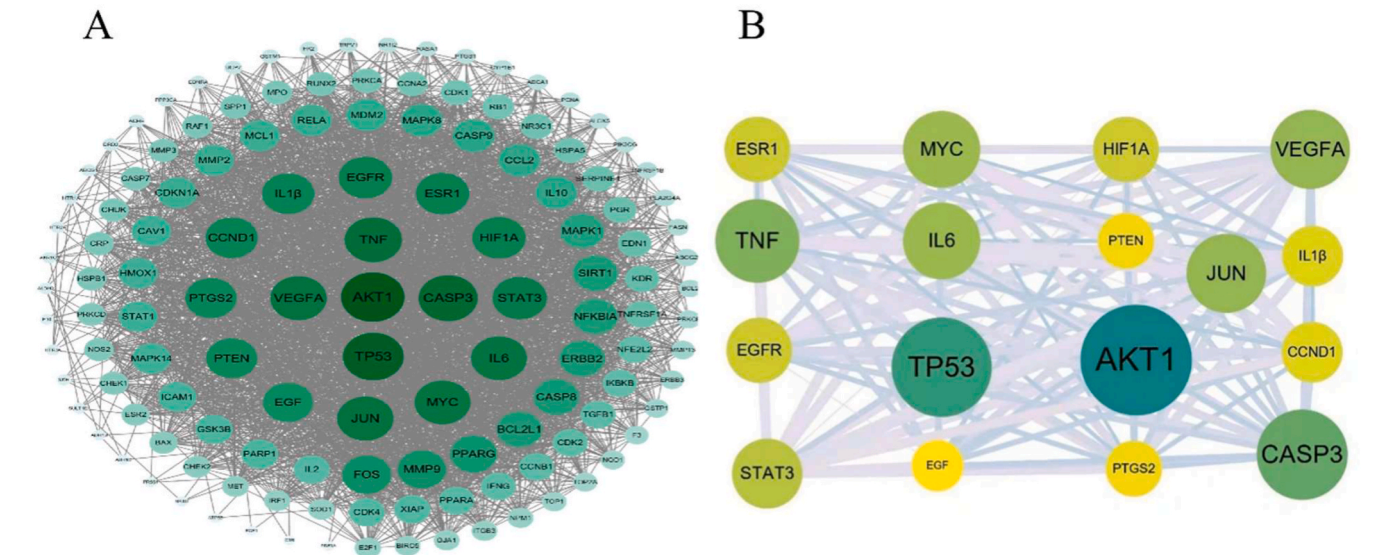


Fig. 3. PPI and core targets. (A) PPI of 134 targets of SSF in alleviating CDDP-induced nephrotoxicity. (B) 17 core targets. The larger the area of the target, the higher the degree value, which is used to evaluate the importance of targets.

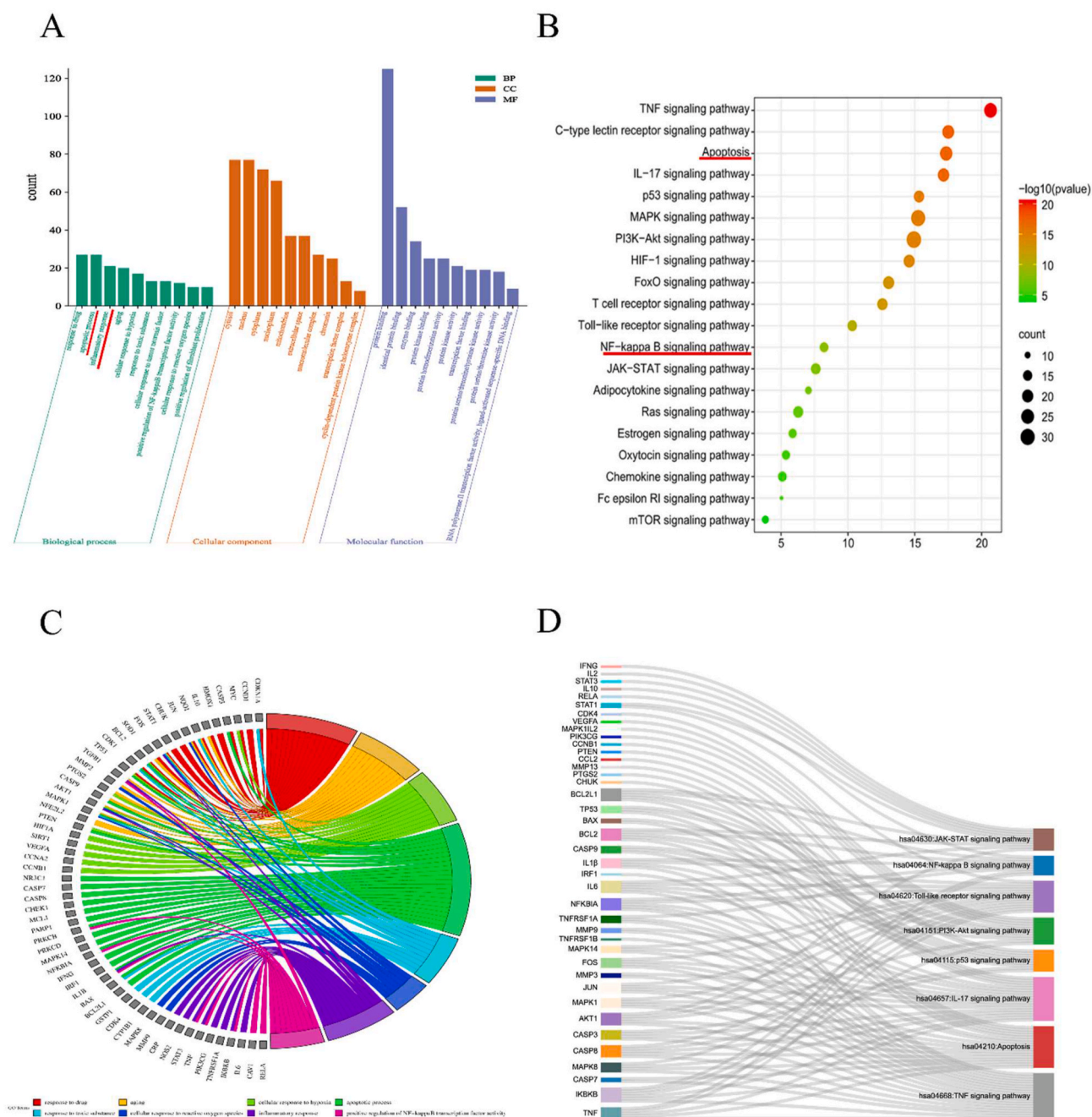


Fig. 4. GO and KEGG enrichment analysis. (A) GO enrichment analysis. The BP terms marked by red lines are related to apoptosis and inflammatory response. (B) KEGG enrichment analysis. Pathways marked by red lines are apoptosis and NF- κ B pathways. (C) The chord diagram of BP enrichment. (D) The chord diagram of KEGG pathway enrichment. (For interpretation of the references to color in this figure legend, the reader is referred to the Web version of this article.)

p65 (Fig. 9B).

4. Discussion

The composition of SSF is based on the famous theory of TCM called “Jun (Monarch)-Chen (Minister) -Zuo (Assistant)-Shi (Courier)”. Radix Astragali nourishes qi of spleen and kidney, hence it is the “Jun”, i.e., the most component of SSF. In addition, studies have reported that Radix Astragali protects kidneys, eliminates urine protein and diuresis. Salvia Miltiorrhiza can activate blood circulation to remove blood stasis; researchers agree that it can protect kidney function and alleviate acute kidney injury caused by ischemia. Rhubarb has defecation activity and can expel turbidity from kidneys. Therefore, Salvia Miltiorrhiza and

Rhubarb play the role of Chen together. Among the other herbs, Alismatis Rhizoma, Poria, and Leonuri Herba can promote diuresis, and Lycii Fructus can tonify kidney and enlighten essence. Together, they play the role of Zuo and Shi.

In this study, Network pharmacology and molecular docking were used to predict the active ingredients, targets, and pathways of SSF to alleviate CDDP nephrotoxicity. Through animal experiments, we confirmed that SSF could indeed suppress inflammation and apoptosis to alleviate CDDP nephrotoxicity. In simple terms, SSF inhibits TLR4/MyD88/NF- κ B to downregulate the expression of IL-6, IL-1 β , TNF- α and, Cox-2 to alleviate inflammation. It also exerts anti-apoptotic effects by upregulating Bcl-2 expression and reducing the activity of Bax, Caspase 3, and Caspase 9. This indicates that SSF is an effective prevention

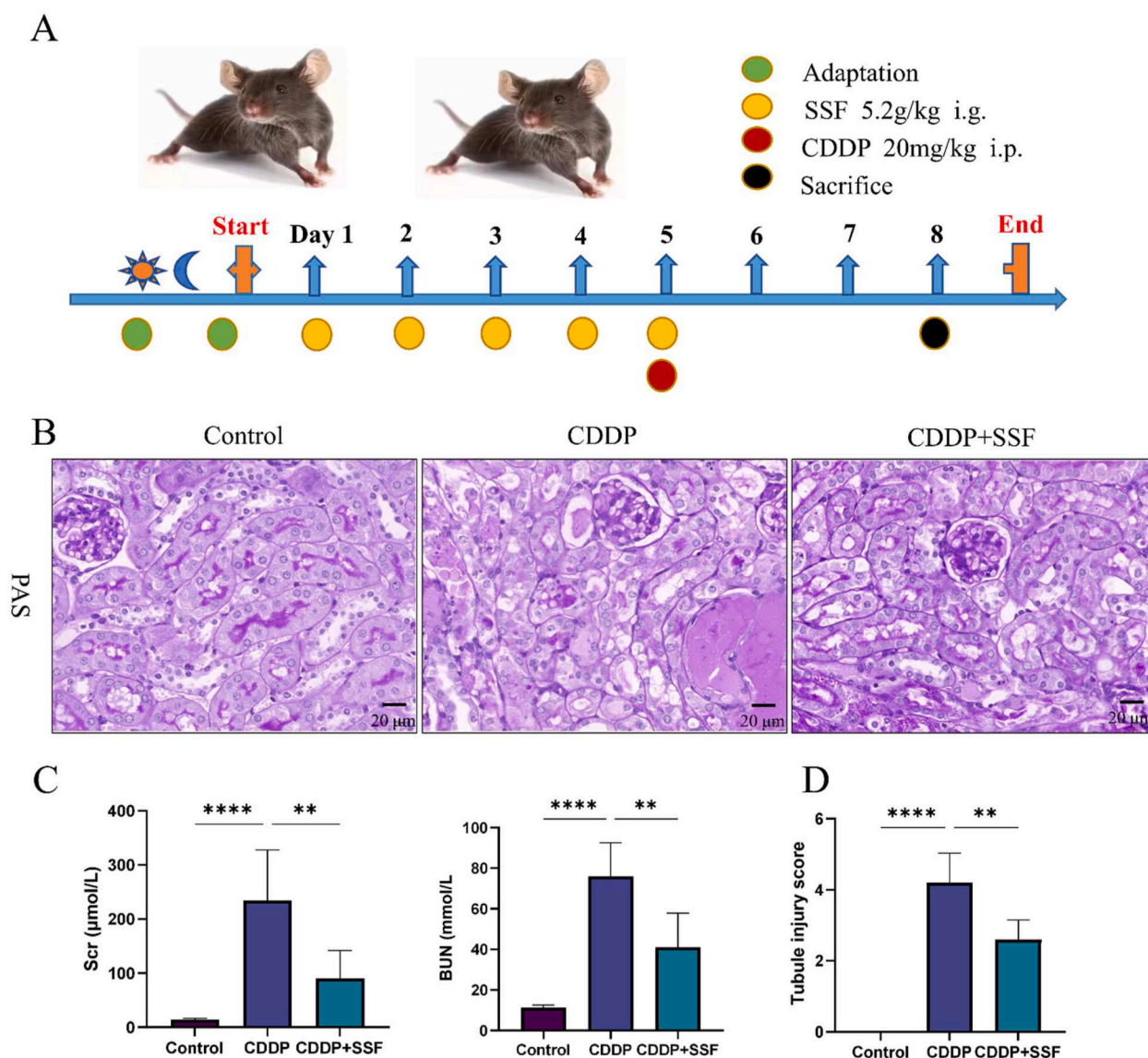


Fig. 5. SSF protected renal tubular and renal function in mice with CDDP-induced nephrotoxicity. (A) Modeling method and drug intervention schedule. (B) Representative images of renal tubular injury by PAS staining (Scale bar:20 μ m). (C) Levels of Scr and BUN of three groups of mice. Control=6, CDDP=6, CDDP+SSF=6. (D) Renal tubular injury scores. Renal tubular injury scores were scored as follows: no damage=0; 0%–10% damage=1; 11%–25% damage=2; 26%–75% damage=3; 76%–100% damage=4. ** P <0.01, **** P <0.0001.

intervention for CDDP-induced nephrotoxicity.

Considering the large amounts of Radix Astragali, Salvia Miltiorrhiza, and Rhubarb in SSF, we separately performed TLC to identify these 3 herbs in SSF. According to Chinese Pharmacopoeia (2020 Edition), the Astragaloside IV content of Radix Astragali must not be less than 0.040%. Astragaloside IV is the best bioactive component of Radix Astragali and one of the major criteria for evaluating the superiority and inferiority of Radix Astragali; thus, we selected Astragaloside IV as the reference standards of Radix Astragali. Salvianic acid A is an exclusive component of Salvia Miltiorrhiza and one of the major pharmacodynamic components, hence Salvianic acid A was selected as the reference standard for determining Salvia Miltiorrhiza. According to Chinese Pharmacopoeia (2020 Edition), the total amount of Rhein, Emodin, Chrysophanol, and Emodin Methyl Ether in Rhubarb must not be less than 1.5%. Therefore, we used Rhein and Rhubarb reference drug as the reference standards for Rhubarb. Astragaloside IV has significant anti-inflammatory and anti-oxidant effects, which can alleviate CDDP-induced nephrotoxicity even AKI, sepsis or IR-induced AKI (Song et al., 2021; Su et al., 2022; Tang et al., 2022; Yan et al., 2017; Zhou

et al., 2017). Rhein can inhibit the NF- κ B pathway (Liu et al., 2021) and promote the hydrolysis of Toll receptor (Bi et al., 2018) to reduce the inflammatory reaction in rats with AKI.

In this regard, we hope to predict more active ingredients of SSF and identify core protein targets. Using network pharmacology, we identified 127 active ingredients of SSF. Further analysis identified 14 active ingredients as the main active ingredients (Table 2), including Flavones, Isoflavones, Anthraquinones, and other types. Previous studies have found that Quinones such as Aloe-emodin can reduce Caspase 3 activity, activate autophagy, and reduce CDDP-induced nephrotoxicity in vitro by regulating the AMPK/TOR pathway (Liu et al., 2016). In addition, some active ingredients can reduce inflammatory response by inhibiting the NF- κ B pathway.

Initially, we identified 134 common targets of SSF and CDDP-induced nephrotoxicity. However, additional analysis revealed 17 of the 134 targets as core targets (Fig. 3B). Among them, AKT1, TP53, and CASP 3 were closely related to apoptosis, whereas TNF- α , IL-6, and IL-1 β were classical factors associated with inflammatory-induced damage. TNF- α can directly induce cells to release a large number of

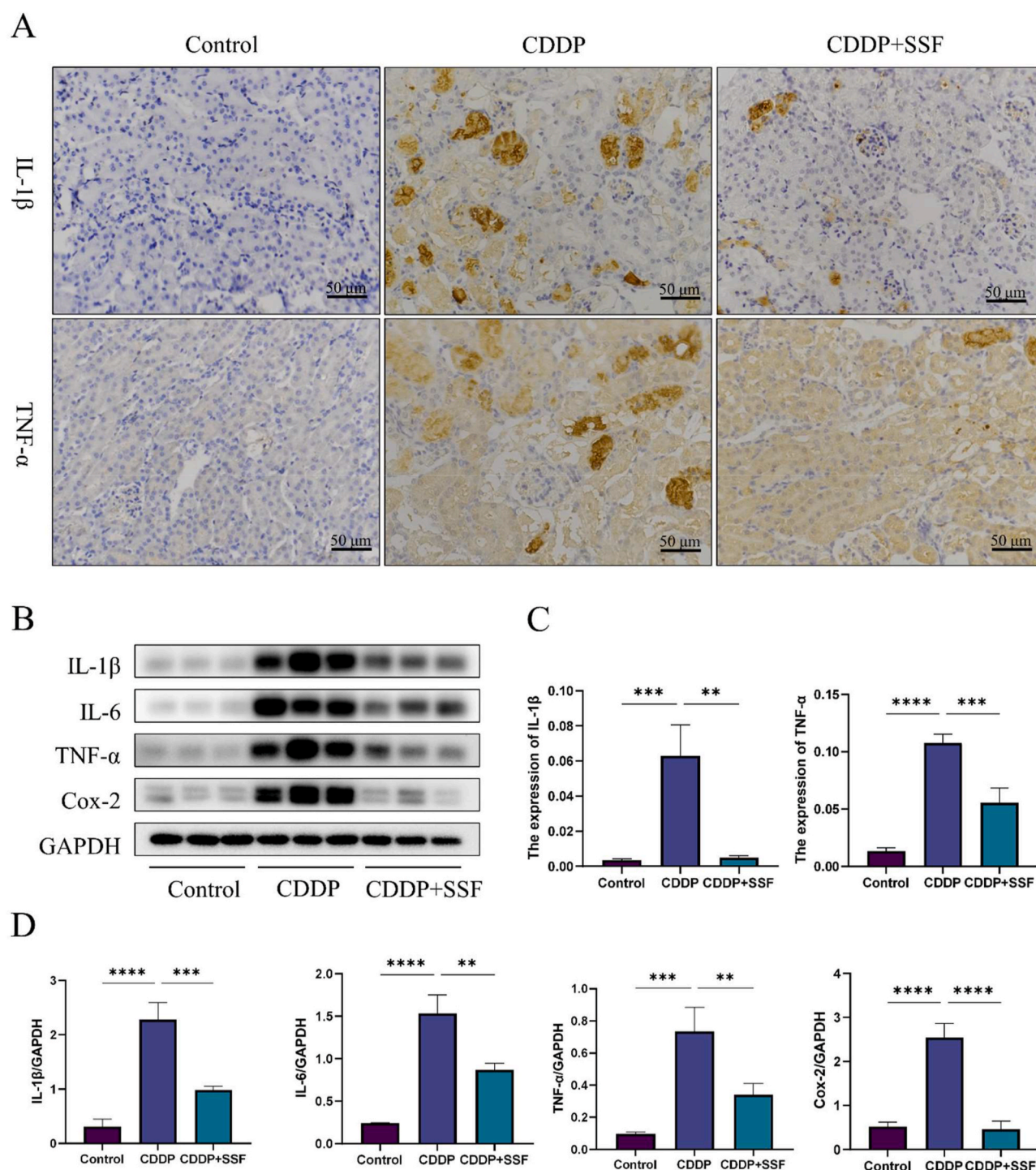


Fig. 6. SSF reduced inflammation in mice with CDDP-induced nephrotoxicity. (A) Representative images of IL-1 β and TNF- α by IHC (Scale bar:50 μ m). (B) The expression of IL-1 β , IL-6, TNF- α , and Cox-2 by WB. (C) The quantitative analysis of IHC for IL-1 β and TNF- α . (D) The quantitative analysis of WB for IL-1 β , IL-6, TNF- α , and Cox-2. CDDP=3, CDDP+SSF=3. ** P <0.01, *** P <0.001, **** P <0.0001.

inflammatory cytokines and inflammatory mediators, thereby secreting many cytokines and the key factors contributing to the systemic inflammatory response (Miller et al., 2010). AKT1 is a member of the protein kinase family. Studies have shown that AKT1 deletion aggravates renal tubular injury and inflammatory response in mice with IR-induced AKI (Kim et al., 2020). Moreover, P53-mediated apoptosis is a key mechanism of CDDP-induced nephrotoxicity (Wu et al., 2021). Through the analysis of core targets, apoptosis and inflammatory response are important factors that should be further investigated.

In subsequent GO enrichment analysis, BP was involved in inflammatory response and apoptosis (Fig. 4A). The KEGG enrichment analysis involved apoptosis and inflammation-related pathways, including Apoptosis, TNF pathway, IL-17, and NF- κ B pathways (Fig. 4B). GO and KEGG enrichment analysis further illustrated that SSF may alleviate CDDP-induced nephrotoxicity by inhibiting apoptosis and inflammation.

Experiments in mice confirmed that intraperitoneal injection of 20 mg/kg of CDDP impair the renal function, manifested as a significant

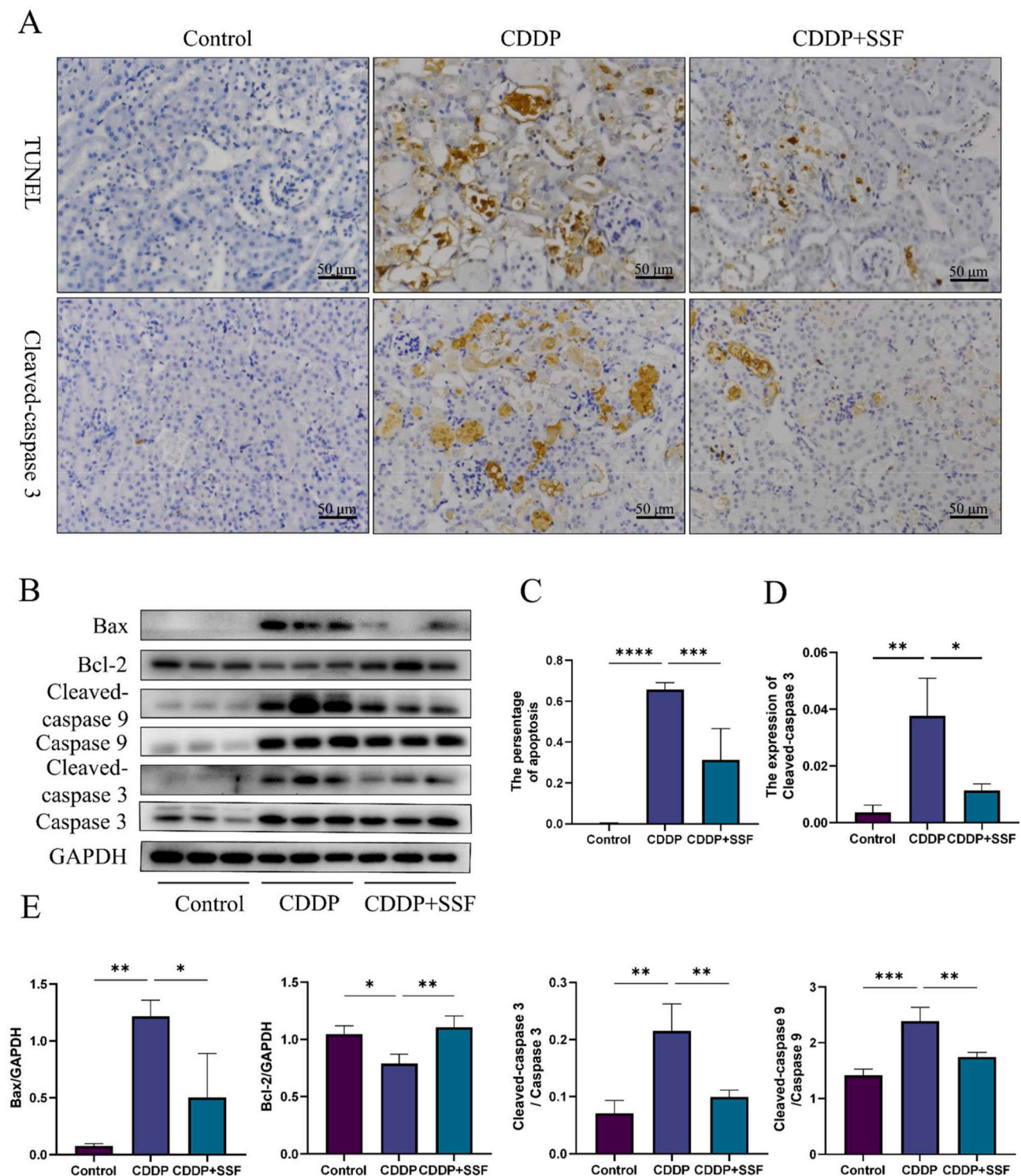


Fig. 7. SSF reduced apoptosis in mice with CDDP-induced nephrotoxicity. (A) Representative images showing apoptosis in renal tubules cells by TUNEL staining and Cleaved-caspase 3 by IHC (Scale bar:50 μ m). (B) The expression of Bax, Bcl-2, Cleaved-caspase 9, Caspase 9, Cleaved-caspase 3 and Caspase 3 by WB. (C) The percentage of apoptosis in renal tubule cells. Apoptosis percentage= (Number of apoptotic cells/Total number of cells) $\times$ 100%. (D) Quantitative analysis of IHC for Cleaved-caspase 3. (E) Quantitative analysis of WB for Bax, Bcl-2, Cleaved-caspase 9/Caspase 9 and Cleaved-caspase 3/Caspase 3. Control=3, CDDP=3, CDDP+SSF=3, * P <0.05, ** P <0.01, *** P <0.001, **** P <0.0001.

increase in Scr and BUN levels (Fig. 5C). In addition, results of PAS staining showed that CDDP treatment induced significant damage the renal tubules in the kidney (Fig. 5B). However, pre-treatment with 5.2 g/kg of SSF per day for 5 consecutive days before administration of CDDP significantly suppressed CDDP-induced damage to the renal

tubules.

In addition, SSF suppressed the inflammatory response. The nuclear transcription factor-kappa B (NF- κ B) pathway is a classical signaling pathway that induces the inflammatory response. Toll-like receptor 4 (TLR4) is an upstream molecule in NF- κ B signaling and an important

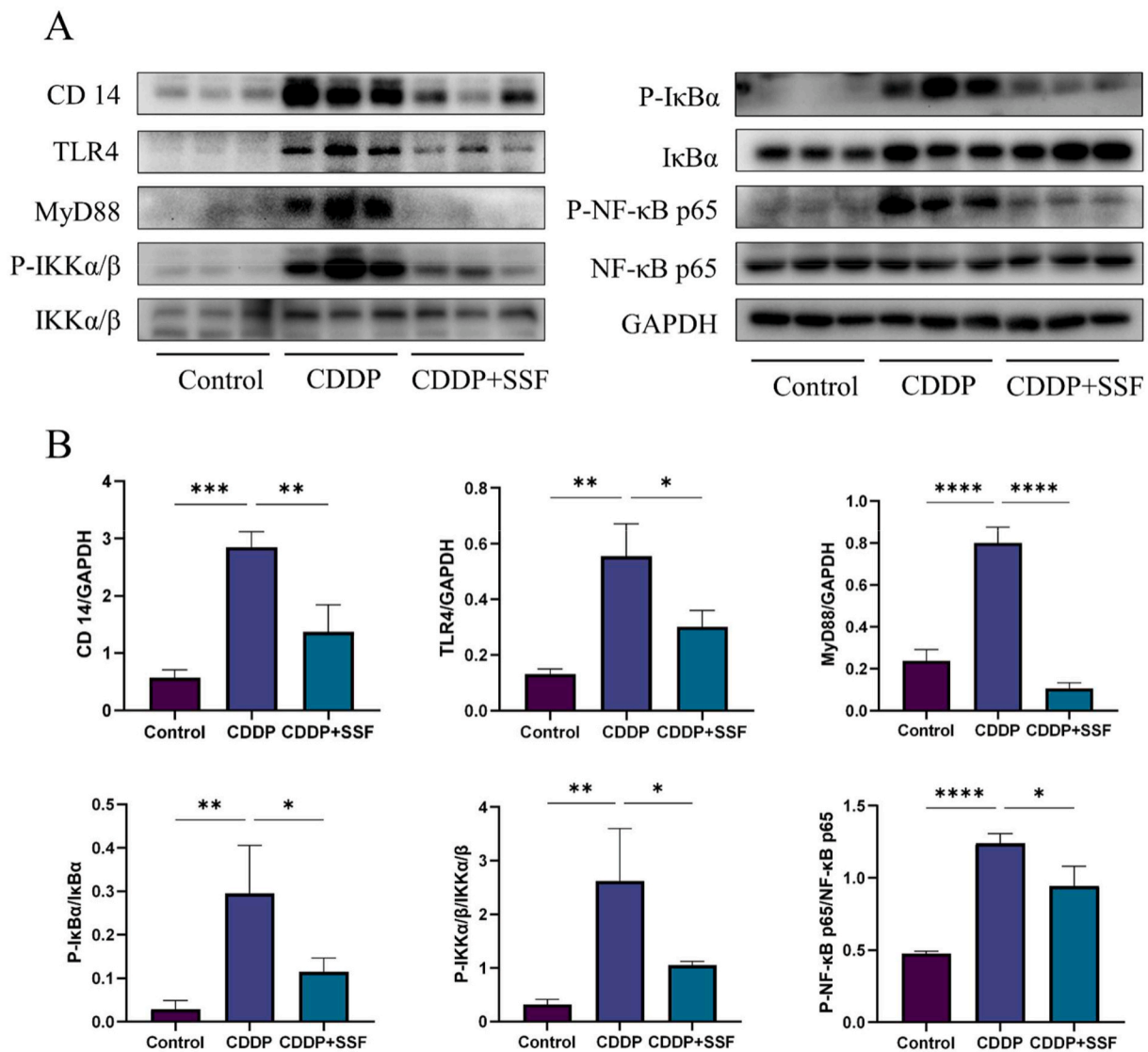


Fig. 8. SSF inhibited the TLR4/MyD88/NF- κ B pathway in mice with CDDP-induced nephrotoxicity. (A) The expression of CD 14, TLR4, MyD88, P-IKK α / β , IKK α / β , P-I κ B α , I κ B α , P-NF- κ B p65 and NF- κ B p65 by WB. (B) The quantitative analysis of WB for TLR4, MyD88, P-IKK α / β /IKK α / β , P-I κ B α /I κ B α , P-NF- κ B p65/NF- κ B p65. Control=3, CDDP=3, CDDP+SSF=3. * P <0.05, ** P <0.01. *** P <0.001, **** P <0.0001.

transmembrane receptor that activates the immune system (Shi et al., 2013). Lipopolysaccharide (LPS) is a major component of the cell outer wall of Gram-negative bacilli. The Cluster of differentiation 14 (CD14) is a member of the cell surface glycoprotein family. The binding of LPS to CD14 can activate TLR4 in monocytes (Luo et al., 2022). TLR4 can initiate intracellular signaling cascades such as MyD88 and NF- κ B (Verstrepen et al., 2008), which activate downstream inflammatory factors (Saleh et al., 2021). Upon binding to the stimulatory factor LPS, TLR4 activates macrophages and recruits MyD88 to initiate a cellular cascade that activates NF- κ B. NF- κ B exists in the inactive form or combined with I κ B α in the normal state. MyD88 first stimulates the activation of IKK α / β , and the activation of IKK α / β reduces the binding of I κ B α to NF- κ B p65, allowing NF- κ B translocation from the cytoplasm to the nucleus where it stimulates and enhances the expression of inflammatory factors including TNF- α , IL-1 β , and IL-6 (Guo et al., 2021). The activation of the NF- κ B pathway has been implicated in the pathogenesis of acute kidney injury (Wang et al., 2020). In our study, unlike mice exposed to CDDP, the TLR4/MyD88/NF- κ B signaling pathway was inhibited in mice pretreated with SSF (Fig. 8A). In addition, the levels of the corresponding inflammatory factors TNF- α , IL-1 β , IL-6, and Cox-2 were lower in mice pretreated with SSF (Fig. 6B), indicating that SSF

effectively alleviated the inflammatory response in the kidney by inhibiting TLR4/MyD88/NF- κ B pathway.

The TLR4/MyD88/NF- κ B pathway also regulates the anti-apoptotic process in AKI. Previous studies have shown that ulinastatin suppresses inflammation and apoptosis by inhibiting the TLR4/NF- κ B pathway in rhabdomyolysis-induced AKI (Chen et al., 2021). NF- κ B activation can up-regulate Bcl-2 expression and down-regulate Bax expression. Bcl-2 is an anti-apoptotic protein that can inhibit Caspase 9 activation by the pro-apoptotic protein Bax (Xing et al., 2019). Apoptosis caused by caspase activation is an important mechanism of CDDP-induced acute kidney injury, which directly causes cell apoptosis and disintegration, thereby disrupting the renal function (Li et al., 2020). In simple terms, intracellular and extracellular stimuli activate and trigger the cleavage of the Caspase 9 promoter; Caspase 9 further activates the downstream executor Caspase 3. The cleavage of Caspase 3 causes a series of cascade effects, resulting in DNA fragmentation, nuclear condensation, and ultimately apoptosis (Salvesen and Riedel, 2008). Caspase-3 is an important apoptosis-related protein downstream of the NF- κ B signaling pathway and the primary effector of apoptosis (Wang et al., 2021). Its activation and cleavage mark the irreversible completion of the apoptotic process (Doherty and Baehrecke, 2018). In

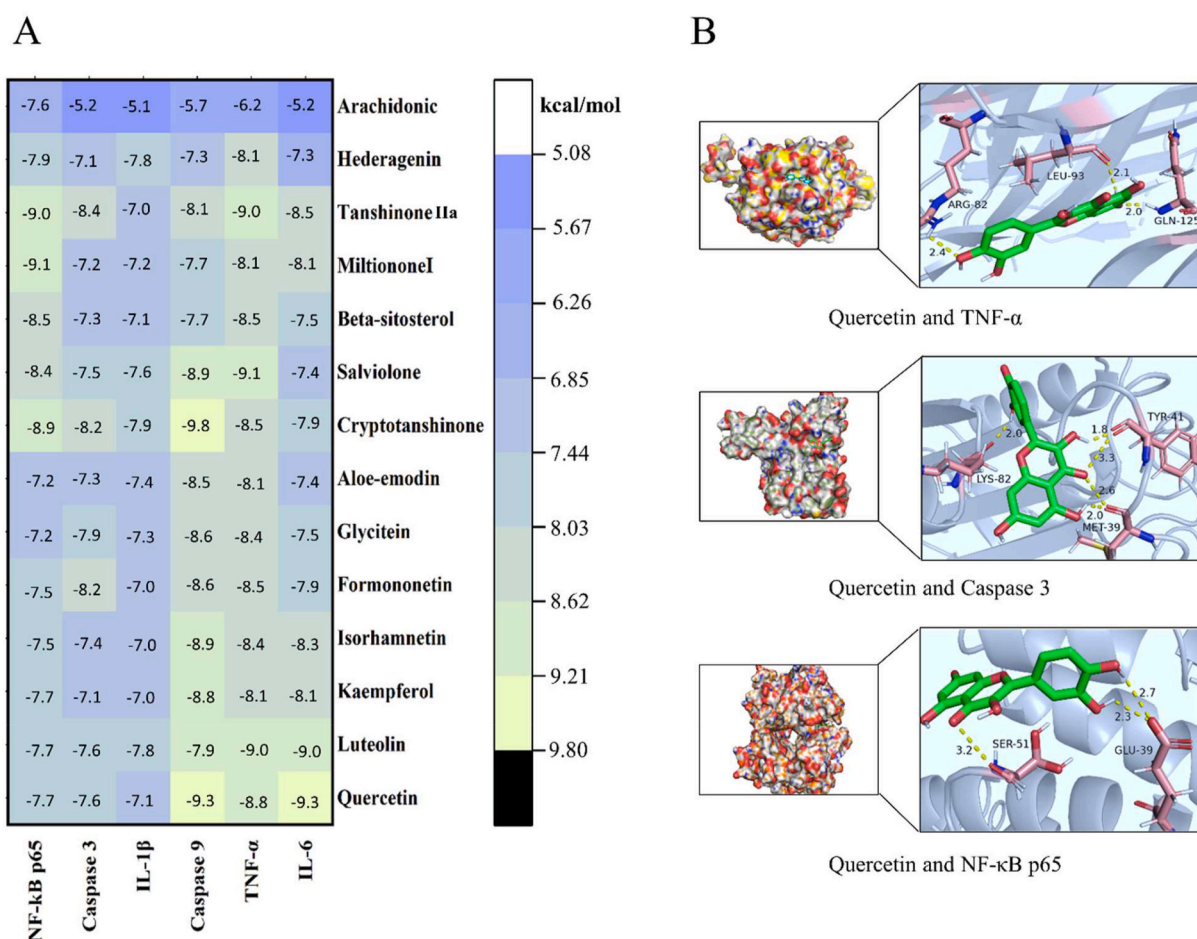


Fig. 9. Molecular docking of main active ingredients and core targets. (A) Heat map of ligand and receptor binding values. (B) The simulated docking effects of Quercetin and TNF-α, Caspase 3, NF-κB p65.

line with previous studies, SSF can inhibit the TLR4/MyD88/NF-κB pathway, upregulate Bcl-2 expression and downregulate the expression of Bax, Cleaved-caspase 3, and Cleaved-caspase 9, thereby alleviating apoptosis of renal tubular cells in a mouse model of CDDP-induced nephrotoxicity (Fig. 7B).

Active ingredients of herbal formulations are diverse, hence modulate multiple targets and pathways. Besides the TLR4/MyD88/NF-κB pathway, the PI3K/AKT, p53, and AMPK pathways may also regulate CDDP-induced nephrotoxicity. Among the 14 active ingredients (Table 2) predicted by network pharmacology, Aloe emodin can inhibit the apoptosis of T24 human bladder cancer cells through the p53 pathway (Lin et al., 2006). Tanshinone IIa could not only downregulate P53-mediated SLC7A11 to inhibit iron death in gastric cancer cells (Guan et al., 2020), but also reduce autophagy in myocardial cells via the AMPK/mTOR pathway (Zhang et al., 2019b). These active ingredients of SSF may play a similar role in the treatment of CDDP nephrotoxicity; therefore, we believe that network pharmacology is an efficient approach for analyzing herbal formulations.

Finally, the 14 main active ingredients of SSF were subjected to molecular docking simulations with the experimentally validated core targets (Fig. 9A). This further indicates that we may have successfully found the chemical substances with pharmacological effects of SSF. The above studies also confirm that network pharmacology can help to reveal the mechanism of action of herbal formulations.

Although SSF has been proven to be effective in the nephrotoxicity of cisplatin in mice, its clinical efficacy remains to be further verified. Besides, additional investigations are necessary since it remains unclear as to which active components in SSF modulate anti-inflammation and

anti-apoptosis. And the target of SSF to alleviate CDDP-induced nephrotoxicity needs further exploration.

5. Conclusion

In conclusion, through network pharmacology, we predicted that SSF can alleviate cisplatin-induced nephrotoxicity through multiple targets and pathways. Furthermore, animal experiments confirmed that SSF can inhibit inflammation and apoptosis, thereby reducing Scr, BUN, and renal pathological damage in the mouse model of cisplatin-induced nephrotoxicity, which may be related to the inhibition of the TLR4/MyD88/NF-κB pathway. Therefore, SSF is a promising kidney protection agent.

Author contributions

Shudong Yang and Jiandong Lu conceived and supervised the research. Xiaoming Jin and Riming He designed, performed, analyzed most of the experiments, and drafted the manuscript. Jiahui Liu, Yuzhi Wang, Zhongtang Li, and Beibei Jiang provided the technical advice. Everyone contributed to the writing.

Fundings

This study was financed by grants from Natural Science Foundation of Guangdong Province of China (2020A1515011151), Traditional Chinese Medicine Bureau of Guangdong Province (20223015), Shenzhen Fund for Guangdong Provincial High Level Clinical Key Specialties,

and Shenzhen Traditional Chinese Medicine Hospital “3030 Program” Chinese Medicine Clinical Research Project (G3030202111).

Declaration of competing interest

The authors declare no known financial or personal conflicts that could affect the work reported in this article.

Data availability

Data will be made available on request.

Acknowledgment

We would like to thank to Dr. Jianping Chen, and Professor Baokun He from Shenzhen Hospital of Traditional Chinese Medicine, and Ms. Bai Rong from Heilongjiang University of Traditional Chinese Medicine for providing us with help.

Appendix A. Supplementary data

Supplementary data to this article can be found online at <https://doi.org/10.1016/j.jep.2023.116168>.

References

- Bailey, C., Vergoten, G., 2020. Glycyrrhizin: an alternative drug for the treatment of COVID-19 infection and the associated respiratory syndrome? *Pharmacol. Therapeut.* 214 <https://doi.org/10.1016/j.pharmthera.2020.107618>.
- Bi, F., Chen, F., Li, Y., Wei, A., Cao, W., 2018. Klotho preservation by Rhein promotes toll-like receptor 4 proteolysis and attenuates lipopolysaccharide-induced acute kidney injury. *J. Mol. Med.* 96 (9), 915–927. <https://doi.org/10.1007/S00109-018-1644-7>.
- Chen, F., Zhu, J., Wang, W., 2021. Ulinastatin attenuates LPS-induced inflammation and inhibits endoplasmic reticulum stress-induced apoptosis in renal tubular epithelial cells via regulation of the TLR4/NF- κ B and Nrf2/HO-1 pathways. *Inflammation* 44 (6), 2323–2332. <https://doi.org/10.1007/S10753-021-01505-Z>.
- Chen, W., Zhou, L., Qiao, Y., Qi, J., Fu, G., Xia, X., 2019. Quality evaluation of Ilex asprella based on simultaneous determination of five bioactive components, chlorogenic acid, Luteoloside, quercetin, quercetin, and kaempferol, using UPLC-Q-TOF MS study. *J. AOAC Int.* 102 (5), 1414–1422. <https://doi.org/10.5740/JAOACINT.18-0391>.
- Crona, D.J., Faso, A., Nishijima, T.F., McGraw, K.A., Galsky, M.D., Milowsky, M.I., 2017. A systematic review of strategies to prevent cisplatin-induced nephrotoxicity. *Oncol.* 22 (5), 609–619. <https://doi.org/10.1634/THEONCOLOGIST.2016-0319>.
- Daina, A., Michielin, O., Zoete, V., 2019. SwissTargetPrediction: updated data and new features for efficient prediction of protein targets of small molecules. *Nucleic Acids Res.* 47 (W1), W357–W364. <https://doi.org/10.1093/nar/gkz382>.
- Dasari, S., Bernard Tchounwou, P., 2014. Cisplatin in cancer therapy: molecular mechanisms of action. *Eur. J. Pharmacol.* 740, 364–378. <https://doi.org/10.1016/J.EJPHAR.2014.07.025>.
- Doherty, J., Baehrecke, E.H., 2018. Life, death and autophagy. *Nat. Cell Biol.* 20 (10), 1110–1117. <https://doi.org/10.1038/S41556-018-0201-5>.
- Dou, J.Y., Zhang, M., Cen, H., Chen, Y.Q., Wu, Y.F., Lu, F., Zhou, J., Liu, X.S., Gu, Y.Y., 2022. Salvia miltiorrhiza bunge (danshen) and bioactive compound tanshinone IIA alleviates cisplatin-induced acute kidney injury through regulating PXR/NF- κ B signaling. *Front. Pharmacol.* 13 <https://doi.org/10.3389/FPHAR.2022.860383>.
- Galgamuwa, R., Hardy, K., Dahlstrom, J.E., Blackburn, A.C., Wium, E., Rooke, M., Cappello, J.Y., Tummala, P., Patel, H.R., Chuah, A., Tian, L., McMorro, L., Board, P. G., Theodoratos, A., 2016. Dichloroacetate prevents cisplatin-induced nephrotoxicity without compromising cisplatin anticancer properties. *J. Am. Soc. Nephrol.* 27 (11), 3331–3344. <https://doi.org/10.1681/ASN.2015070827/-DCS SUPPLEMENTAL>.
- Gu, X., Zhang, G., Wang, Q., Song, J., Li, Y., Xia, C., Zhang, T., Yang, L., Sun, J., Zhou, M., 2022. Integrated network pharmacology and hepatic metabolomics to reveal the mechanism of Acanthopanax senticosus against major depressive disorder. *Front. Cell Dev. Biol.* 10 <https://doi.org/10.3389/FCELL.2022.900637>.
- Guan, Z., Chen, J., Li, X., Dong, N., 2020. Tanshinone IIA induces ferroptosis in gastric cancer cells through p53-mediated SLC7A11 down-regulation. *Biosci. Rep.* 40 (8) <https://doi.org/10.1042/BSR20201807>.
- Guo, S., Li, W., Chen, F., Yang, S., Huang, Y., Tian, Y., Xu, D., Cao, N., 2021. Polysaccharide of Atractylodes macrocephala Koidz regulates LPS-mediated mouse hepatitis through the TLR4-MyD88-NF κ B signaling pathway. *Int. Immunopharm.* 98 <https://doi.org/10.1016/J.INTIMP.2021.107692>.
- Hamano, H., Ikeda, Y., Tsuchiya, K., 2021. Diphenhydramine may be a preventive medicine against cisplatin-induced kidney toxicity. *Kidney Int.* 99 (4), 885–899. <https://doi.org/10.1016/j.kint.2020.10.041>.
- Hao, Y., Miao, J., Liu, W., Peng, L., Chen, Y., Zhong, Q., 2021. Formononetin protects against cisplatin-induced acute kidney injury through activation of the PPAR α /Nrf2/HO-1/NQO1 pathway. *Int. J. Mol. Med.* 47 (2), 511–522. <https://doi.org/10.3892/IJMM.2020.4805/DOWNLOAD>.
- He, R., Liu, J., Chen, Y., Liao, Y., Wang, Y., Jin, X., Li, Z., Liu, S., Lu, J., Yang, S., 2022. A Chinese medicine compound alleviates cisplatin-induced acute kidney injury via its antiapoptosis and anti-inflammation effects in mice. *Evid. base Compl. Alternative Med.: ECAM* 1–10. <https://doi.org/10.1155/2022/7841284>, 2022.
- Huang, D.W., Sherman, B.T., Lempicki, R.A., 2009. Systematic and integrative analysis of large gene lists using DAVID bioinformatics resources. *Nat. Protoc.* 4 (1), 44–57. <https://doi.org/10.1038/nprot.2008.211>.
- Kim, I.Y., Park, Y.K., Song, S.H., Seong, E.Y., Lee, D.W., Bae, S.S., Lee, S.B., 2020. Akt1 is involved in tubular apoptosis and inflammatory response during renal ischemia-reperfusion injury. *Mol. Biol. Rep.* 47 (12), 9511–9520. <https://doi.org/10.1007/S11033-020-06021-1>.
- Li, Q., Yan, X.T., Zhao, L.C., En, S., He, Y.F., Liu, W.C., Wang, Z., Li, X.D., Jiang, S., Li, W., 2020. α -Mangostin, a dietary xanthone, exerts protective effects on cisplatin-induced renal injury via PI3K/akt and JNK signaling pathways in HEK293 cells. *ACS Omega* 5 (32), 19960–19967. <https://doi.org/10.1021/ACSOMEGA.0C01121>.
- Lin, J.G., Chen, G.W., Li, T.M., Chouh, S.T., Tan, T.W., Chung, J.G., 2006. Aloe-emodin induces apoptosis in T24 human bladder cancer cells through the p53 dependent apoptotic pathway. *J. Urol.* 175 (1), 343–347. [https://doi.org/10.1016/S0022-5347\(05\)00005-4](https://doi.org/10.1016/S0022-5347(05)00005-4).
- Liu, H., Gu, L.B., Tu, Y., Hu, H., Huang, Y.R., Sun, W., 2016. Emodin ameliorates cisplatin-induced apoptosis of rat renal tubular cells in vitro by activating autophagy. *Acta Pharmacol. Sin.* 37 (2), 235–245. <https://doi.org/10.1038/APS.2015.114>.
- Liu, M., Wang, L., Wu, X., Gao, K., Wang, F., Cui, J., Zhao, J., Peng, L., Wang, J., Jia, Y., Wen, A., 2021. Rhein protects 5/6 nephrectomized rat against renal injury by reducing inflammation via NF- κ B signaling. *Int. Urol. Nephrol.* 53 (7), 1473–1482. <https://doi.org/10.1007/S11255-020-02739-W>.
- Li, Y., Liu, J.&X., 2011. Study on the quality standard of Shenshuifu granules. *Hebei J. Tradit. Chin. Med.* 43–44, 02. <https://doi.org/10.16370/j.cnki.13-1214/r.2011.02.032>.
- Luo, J., Lu, J., Zeng, J., Ma, Y., Gong, Q., Wang, Z., Zhang, X., Quan, J., 2022. Single-cell RNA analysis of chemokine expression in heterogeneous CD14 + monocytes with lipopolysaccharide-induced bone resorption. *Exp. Cell Res.* 113343 <https://doi.org/10.1016/J.YEXCR.2022.113343>.
- Miller, R.P., Tadagavadi, R.K., Ramesh, G., Reeves, W.B., 2010. Mechanisms of cisplatin nephrotoxicity. *Toxins* 2 (11), 2490–2518. <https://doi.org/10.3390/TOXINS2112490>.
- Nair, A., Morsy, M.A., Jacob, S., 2018. Dose translation between laboratory animals and human in preclinical and clinical phases of drug development. In: *Drug Development Research*, vol. 79. Wiley-Liss Inc, pp. 373–382. <https://doi.org/10.1002/ddr.21461>. Issue 8.
- Ozkok, A., Edelstein, C.L., 2014. Pathophysiology of cisplatin-induced acute kidney injury. *BioMed Res. Int.* <https://doi.org/10.1155/2014/967826>, 2014.
- Qi, L., Luo, Q., Zhang, Y., Jia, F., Zhao, Y., Wang, F., 2019. Advances in toxicological research of the anticancer drug cisplatin. *Chem. Res. Toxicol.* 32 (8), 1469–1486. <https://doi.org/10.1021/ACS.CHEMRESTOX.9B00204>.
- Ramesh, G., Reeves, W.B., 2004. Inflammatory cytokines in acute renal failure. *Kidney Int.* 66 (91) <https://doi.org/10.1111/J.1523-1755.2004.09109.X>.
- Ru, J., Li, P., Wang, J., Zhou, W., Li, B., Huang, C., Li, P., Guo, Z., Tao, W., Yang, Y., Xu, X., Li, Y., Wang, Y., Yang, L., 2014. TCMSP: a database of systems pharmacology for drug discovery from herbal medicines. *J. Cheminf.* 6 (1) <https://doi.org/10.1186/1758-2946-6-13>.
- Safran, M., Rosen, N., Twik, M., BarShir, R., Stein, T.I., Dahary, D., Fishilevich, S., Lancet, D., 2021. The GeneCards suite. In: *Practical Guide to Life Science Databases*. Springer Nature Singapore, pp. 27–56. https://doi.org/10.1007/978-981-16-5812-9_2.
- Saleh, H.A., Yousef, M.H., Abdelnaser, A., 2021. The anti-inflammatory properties of phytochemicals and their effects on epigenetic mechanisms involved in TLR4/NF- κ B-Mediated inflammation. *Front. Immunol.* 12 <https://doi.org/10.3389/FIMMU.2021.606069/PDF>.
- Salvesen, G.S., Riedl, S.J., 2008. Caspase mechanisms. *Adv. Exp. Med. Biol.* 615, 13–23. https://doi.org/10.1007/978-1-4020-6554-5_2.
- Santos, N. A. G. D., Ferreira, R.S., Santos, A. C. D., 2020. Overview of cisplatin-induced neurotoxicity and ototoxicity, and the protective agents. *Food Chem. Toxicol.* 136 <https://doi.org/10.1016/J.FCT.2019.111079>.
- Shi, H., Dong, L., Jiang, J., Zhao, J., Zhao, G., Dang, X., Lu, X., Jia, M., 2013. Chlorogenic acid reduces liver inflammation and fibrosis through inhibition of toll-like receptor 4 signaling pathway. *Toxicology* 303, 107–114. <https://doi.org/10.1016/J.TOX.2012.10.025>.
- Singh, B.K., Chaudhari, A.K., Das, S., Tiwari, S., Maurya, A., Singh, V.K., Dubey, N.K., 2022. Chitosan encompassed Aniba roseoedora essential oil as innovative green candidate for antifungal and antiflatulogenic activity in millets with emphasis on cellular and its mode of action. *Front. Microbiol.* 13 <https://doi.org/10.3389/FMICB.2022.970670>.
- Song, Y., Hu, T., Gao, H., Zhai, J., Gong, J., Zhang, Y., Tao, L., Sun, J., Li, Z., Qu, X., 2021. Altered metabolic profiles and biomarkers associated with astragaloside IV-mediated protection against cisplatin-induced acute kidney injury in rats: an HPLC-TOF/MS-based untargeted metabolomics study. *Biochem. Pharmacol.* 183 <https://doi.org/10.1016/J.BCP.2020.114299>.
- Su, Y., Xu, J., Chen, S., Feng, J., Li, J., Lei, Z., Qiao, L., Wang, Y., Zeng, D., 2022. Astragaloside IV protects against ischemia/reperfusion (I/R)-induced kidney injury based on the Keap1-Nrf2/ARE signaling pathway. *Transl. Androl. Urol.* 11 (8), 1177–1188. <https://doi.org/10.21037/TAU-22-505/COIF>.

- Szklarczyk, D., Gable, A.L., Nastou, K.C., Lyon, D., Kirsch, R., Pyysalo, S., Doncheva, N. T., Legeay, M., Fang, T., Bork, P., Jensen, L.J., von Mering, C., 2021. The STRING database in 2021: customizable protein-protein networks, and functional characterization of user-uploaded gene/measurement sets. *Nucleic Acids Res.* 49 (D1), D605–D612. <https://doi.org/10.1093/nar/gkaa1074>.
- Tan, R.Z., Wang, C., Deng, C., Zhong, X., Yan, Y., Luo, Y., Lan, H.Y., He, T., Wang, L., 2020. Quercetin protects against cisplatin-induced acute kidney injury by inhibiting Mincle/Syk/NF- κ B signaling maintained macrophage inflammation. *Phytother. Res.* 34 (1), 139–152. <https://doi.org/10.1002/PTR.6507>.
- Tang, J.L., Xin, M., Zhang, L.C., 2022. Protective effect of Astragalus membranaceus and Astragaloside IV in sepsis-induced acute kidney injury. *Aging* 14 (14), 5855–5877. <https://doi.org/10.18632/AGING.204189>.
- Verstrepen, L., Bekaert, T., Chau, T.L., Tavernier, J., Chariot, A., Beyaert, R., 2008. TLR-4, IL-1R and TNF-R signaling to NF-kappaB: variations on a common theme. *Cell. Mol. Life Sci.* : CMLS 65 (19), 2964–2978. <https://doi.org/10.1007/S00018-008-8064-8>.
- Vijayan, A., 2021. Tackling AKI: prevention, timing of dialysis and follow-up. *Nat. Rev. Nephrol.* 17 (2), 87–88. <https://doi.org/10.1038/S41581-020-00390-3>.
- Wang, P.T., Li, N., Wang, X.Y., Chen, J.I., Geng, C.H., Liu, Z.Q., Fan, H.J., Lv, Q., Hou, S. K., Gong, Y.H., 2021. RIG-I, a novel DAMPs sensor for myoglobin activates NF- κ B/caspase-3 signaling in CS-AKI model. *Military Medical Research* 8 (1). <https://doi.org/10.1186/S40779-021-00333-4>.
- Wang, X., Zhou, J., Yang, J., Wang, S., Yang, L., 2020. Role of TLR4/MyD88/NF- κ B signaling in the contrast-induced injury of renal tubular epithelial cells. *Exp. Ther. Med.* 20 (5) <https://doi.org/10.3892/ETM.2020.9243>, 1–1.
- Wu, W., Fu, Y., Liu, Z., Shu, S., Wang, Y., Tang, C., Cai, J., Dong, Z., 2021. NAM protects against cisplatin-induced acute kidney injury by suppressing the PARP1/p53 pathway. *Toxicol. Appl. Pharmacol.* 418 <https://doi.org/10.1016/J.TAAP.2021.115492>.
- Xing, J. jing, Hou, J. gang, Ma, Z. na, Wang, Z., Ren, S., Wang, Y. ping, Liu, W. cong, Chen, C., Li, W., 2019. Ginsenoside Rb3 provides protective effects against cisplatin-induced nephrotoxicity via regulation of AMPK-/mTOR-mediated autophagy and inhibition of apoptosis in vitro and in vivo. *Cell Prolif* 52 (4). <https://doi.org/10.1111/cpr.12627>.
- Xu, Y., Ma, H., Shao, J., Wu, J., Zhou, L., Zhang, Z., Wang, Y., Huang, Z., Ren, J., Liu, S., Chen, X., Han, J., 2015. A role for tubular necroptosis in cisplatin-induced AKI. *J. Am. Soc. Nephrol.: JASN* 26 (11), 2647–2658. <https://doi.org/10.1681/ASN.2014080741>.
- Yan, W., Xu, Y., Yuan, Y., Tian, L., Wang, Q., Xie, Y., Shao, X., Zhang, M., Ni, Z., Mou, S., 2017. Renoprotective mechanisms of Astragaloside IV in cisplatin-induced acute kidney injury. *Free Radic. Res.* 51 (7–8), 669–683. <https://doi.org/10.1080/10715762.2017.1361532>.
- Yang, L., Brooks, C.R., Xiao, S., Sabbiseti, V., Yeung, M.Y., Hsiao, L.L., Ichimura, T., Kuchroo, V., Bonventre, J.v., 2015. KIM-1-mediated phagocytosis reduces acute injury to the kidney. *J. Clin. Investig.* 125 (4), 1620–1636. <https://doi.org/10.1172/JCI75417>.
- Yang, Q., Yi, Y., 2012. Efficacy of integrated traditional Chinese and Western medicine in the treatment of chronic renal failure. *Shanxi.Tradit. Chin. Med.* 31–32, 03.
- Zhang, N., Cao, J., 2020. Analysis of the clinical effect of Shenshuifu Granule on chronic renal failure. *Hubei.J. Tradit. Chin. Med.* 41–43, 04.
- Zhang, R., Zhu, X., Bai, H., Ning, K., 2019a. Network pharmacology databases for traditional Chinese medicine: review and assessment. *Front. Pharmacol.* 10 (February) <https://doi.org/10.3389/FPHAR.2019.00123>.
- Zhang, X., Wang, Q., Wang, X., Chen, X., Shao, M., Zhang, Q., Guo, D., Wu, Y., Li, C., Wang, W., Wang, Y., 2019b. Tanshinone IIA protects against heart failure post-myocardial infarction via AMPKs/mTOR-dependent autophagy pathway. *Biomedicine & Pharmacotherapy = Biomedecine & Pharmacotherapie* 112. <https://doi.org/10.1016/J.BIOPHA.2019.108599>.
- Zhou, W., Chen, Y., Zhang, X., 2017. Astragaloside IV alleviates lipopolysaccharide-induced acute kidney injury through down-regulating cytokines, CCR5 and p-ERK, and elevating anti-oxidative ability. *Med. Sci. Mon. Int. Med. J. Exp. Clin. Res.: Int. Med. J.Exp.Clin. Res.* 23, 1413–1420. <https://doi.org/10.12659/MSM.899618>.